MODULE 04 · DOMAIN 1 - ARCHITECTURE BUILT FOR THE CLOUD

Service Mesh and Ingress Design

How traffic gets in, and how services find and trust each other once it is inside.

DOMAIN 1 of 5 EXAM WEIGHT 25% MODULE 4 of 52

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North-south traffic needs a front door. East-west traffic between services needs discovery, retries and trust.

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The services and the direction they serve

SERVICE	DIRECTION	BEST AT
Application Load Balancer	North-south	HTTP routing, auth, WAF
Network Load Balancer	North-south	TCP and UDP, static IPs, PrivateLink
Amazon API Gateway	North-south	API keys, quotas, transforms
AWS Load Balancer Controller	North-south	ALB and NLB for Kubernetes
ECS Service Connect	East-west	ECS service discovery and retries
Amazon VPC Lattice	East-west	Any compute, across VPCs and accounts

ON THE EXAM

AWS App Mesh is not on this list. It was discontinued on 30 September 2026.

Application or Network Load Balancer?

CHOOSE ALB WHEN

Layer 7 decisions

- Route on path, host or header
- Authenticate users with OIDC or Cognito
- AWS WAF on the listener
- Lambda functions as targets

CHOOSE NLB WHEN

Layer 4 performance

- TCP, UDP or TLS passthrough
- Static IP or Elastic IP per AZ
- Client source IP preserved
- Front end for a PrivateLink service

THE TIEBREAKER

Fixed IPs for a partner allow-list, non-HTTP protocols, or PrivateLink all mean NLB.

API Gateway: REST API or HTTP API?

CHOOSE REST API WHEN

Full API management

- API keys and usage plans per client
- AWS WAF and resource policies
- Caching and request validation
- Private or edge-optimised endpoints

CHOOSE HTTP API WHEN

Lower cost and latency

- Native JWT authorisers
- Private integration through Cloud Map
- Automatic deployments
- Features the API does not need are absent

THE TIEBREAKER

Per-client quotas, WAF, caching or a private endpoint rule out HTTP APIs. Otherwise HTTP is cheaper.

Kubernetes ingress on EKS

- **AWS Load Balancer Controller** watches the cluster and provisions load balancers.
- **Ingress resource** becomes an ALB; **Service of type LoadBalancer** becomes an NLB.
- **IP target mode** sends traffic straight to pod IPs. Required for Fargate pods.
- **IngressGroup** lets several Ingress resources share one ALB.
- **EKS Auto Mode** runs this capability for you.

> ON THE EXAM
>
> One ALB per Ingress is the default and gets expensive. IngressGroup shares one across teams.

A service mesh gives every call between services **discovery, retries, encryption and metrics** without changing the code.

ECS Service Connect or VPC Lattice?

CHOOSE SERVICE CONNECT WHEN

ECS services together

- All callers and callees run on ECS
- Proxy added to each task automatically
- Retries, outlier detection, free metrics
- Namespace can be shared with RAM

CHOOSE VPC LATTICE WHEN

Any compute, any VPC

- Mixes EC2, ECS, EKS, Lambda and IPs
- Crosses VPCs and accounts, no peering
- IAM auth policies on each request
- No sidecars to run or upgrade

THE TIEBREAKER

Mixed compute, crossing VPCs, or per-request IAM authorisation means Lattice. ECS-only means Service Connect.

VPC Lattice building blocks

COMPONENT	WHAT IT IS
Service network	Logical boundary that VPCs and services join
Service	A named endpoint with listeners and routing rules
Target group	EC2, IP, Lambda, ALB, ECS or Kubernetes targets
Auth policy	IAM policy on the network, the service, or both
Resource configuration	TCP access to resources such as an RDS database
RAM share	How other accounts join the network or use a service

ON THE EXAM

Services join a service network; VPCs associate with it. Neither side needs a route to the other.

VPC Lattice or Transit Gateway?

TRANSIT GATEWAY

Connect networks

- Routes whole CIDR ranges
- Needs non-overlapping addresses
- Any port, any protocol
- Security groups and NACLs do the filtering

VPC LATTICE

Connect services

- Exposes only named services
- Overlapping CIDRs are fine
- HTTP, HTTPS and gRPC; TCP via resources
- IAM decides who may call what

THE TIEBREAKER

Network reachability is Transit Gateway. Service-to-service access with identity is Lattice.

Lattice beyond HTTP: resource configurations

- **Resource configuration** exposes a TCP resource to other VPCs and accounts.
- **By RDS ARN,** by IP address, by domain name, or by CIDR range.
- **Resource gateway:** network interfaces in the provider VPC that receive the traffic.
- **Shared with RAM;** reached via a service network or a resource endpoint.
- **TCP only,** and block transitive sharing so a consumer cannot re-share it.

> ON THE EXAM
>
> A database in another account, no load balancer in front, overlapping CIDRs: a Lattice resource configuration.

Reading the scenario

WHEN THE QUESTION SAYS	IT IS TESTING
Partners allow-list our IP addresses	NLB with Elastic IPs
Different quota for each API client	REST API usage plans
App Mesh in the answer	A discontinued service
ECS only, discovery and retries	ECS Service Connect
Across accounts, overlapping CIDRs	VPC Lattice
Database in another account, privately	Lattice resource configuration

ON THE EXAM

Front door or between services? Decide that first and half the options disappear.

> Meridian Freight exposes a shipment-status service to logistics partners over a proprietary TCP protocol. Several partners' firewalls only permit outbound traffic to IP addresses they have explicitly allow-listed.
>
> The service must see each partner's real source IP address for auditing.

Which front end should the solutions architect use?

A. An Application Load Balancer with AWS WAF

B. Amazon CloudFront in front of an Application Load Balancer

C. An API Gateway Regional REST API

D. A Network Load Balancer with an Elastic IP address in each Availability Zone

◆ PAUSE AND DECIDE

Which front end should the solutions architect use?

A ALB with WAF

HTTP only, and its IP addresses change, so partners cannot allow-list them.

B CloudFront and ALB

Still HTTP only, and edge addresses are many and changing.

C API Gateway REST API

HTTP-based, with no fixed IP addresses.

D NLB with Elastic IPs

Carries raw TCP, fixed addresses, and preserves the client source IP.

THE SIGNAL

A proprietary TCP protocol removes every HTTP option. Allow-listed IPs confirm the NLB.

> Meridian Freight is publishing a rate-quote API to 200 partner companies. Each partner has a contracted monthly request quota and a per-second throttle, and the security team requires AWS WAF in front of the API.
>
> The backend is a set of Lambda functions.

Which solution meets these requirements?

A. An HTTP API with a JWT authoriser for each partner

B. A REST API with an API key and usage plan per partner, and AWS WAF associated with the stage

C. An Application Load Balancer with Lambda targets and AWS WAF

D. An HTTP API behind Amazon CloudFront, with AWS WAF on the distribution

◆ PAUSE AND DECIDE

Which solution meets these requirements?

A. HTTP API with JWT

Authenticates partners, but HTTP APIs have no usage plans, so no per-partner quotas.

B. REST API, usage plans, WAF

Usage plans hold each partner's quota and throttle; WAF attaches to the stage.

C. ALB with WAF

No concept of per-client quotas at all.

D. HTTP API behind CloudFront

Solves WAF by adding CloudFront, but quotas are still missing.

THE SIGNAL

A quota and throttle per client is a usage plan, and usage plans exist only on REST APIs.

Meridian Freight ran its EKS microservices on AWS App Mesh until the service was discontinued. The services run in three AWS accounts, and several also call Lambda functions.

The security team wants every service-to-service call authorised against IAM, and the platform team wants no sidecar proxies to maintain.

What should replace App Mesh?

A. A VPC Lattice service network shared through AWS RAM, with auth policies on each service

B. ECS Service Connect with a shared AWS Cloud Map namespace

C. A self-managed Istio installation on each cluster

D. A Transit Gateway connecting the three VPCs, with security groups between services

◆ PAUSE AND DECIDE

What should replace App Mesh?

A VPC Lattice with auth policies

EKS and Lambda targets, across accounts, IAM on every call, no sidecars.

B ECS Service Connect

For ECS services only. These run on EKS.

C Self-managed Istio

Works, but it is all sidecars and all yours to run and upgrade.

D Transit Gateway

Network reachability only. Security groups cannot authorise against IAM.

THE SIGNAL

EKS plus Lambda, three accounts, IAM per call, no sidecars. Each detail independently points to Lattice.

> Meridian Freight runs 25 ECS services in one account and Region. Each service currently has its own internal Application Load Balancer so that other services can reach it.
>
> The company wants service discovery, automatic retries and per-service traffic metrics, and to remove the internal load balancers, with the LEAST operational overhead and cost.

What should the solutions architect implement?

A. AWS Cloud Map DNS-based service discovery with no proxy

B. Keep the internal load balancers and add AWS X-Ray tracing

C. ECS Service Connect in a single namespace

D. A VPC Lattice service network with each ECS service registered

◆ PAUSE AND DECIDE

What should the solutions architect implement?

A Cloud Map DNS only

Discovery, but no retries and no traffic metrics. Clients see stale DNS on scale-in.

B Keep ALBs, add X-Ray

Keeps 25 load balancers, which the company wants to remove.

C ECS Service Connect

Discovery, retries and free CloudWatch metrics for ECS, with no load balancers.

D VPC Lattice

Works, but adds per-service and per-request charges for reach it does not need.

THE SIGNAL

One account, one Region, ECS only. Nothing here needs Lattice's reach, so the cheaper ECS-native option wins.

> Meridian Freight acquired a regional carrier. Both companies' VPCs use 10.0.0.0/16, and re-addressing either is not possible this year.
>
> Meridian's billing service, on Lambda, must call the carrier's inventory API, on EKS. Only the billing role may call it, and nothing else in either network may be reachable.

Which solution meets these requirements?

A VPC peering between the two VPCs

B A Transit Gateway with both VPCs attached

C A PrivateLink endpoint service in front of a Network Load Balancer

D A VPC Lattice service network, with an auth policy allowing only the billing role

◆ PAUSE AND DECIDE

Which solution meets these requirements?

A VPC peering

Peering cannot connect VPCs with overlapping CIDR ranges.

B Transit Gateway

Overlapping ranges cannot be routed, and it would expose whole networks.

C PrivateLink with an NLB

Handles the overlap, but cannot authorise each call against a specific IAM role.

D VPC Lattice with auth policy

Overlap is irrelevant, only the one service is exposed, and IAM decides who calls.

THE SIGNAL

Overlapping CIDRs remove routing answers. Authorising one IAM role per call removes PrivateLink.

> Analytics jobs on EKS in Meridian Freight's analytics account must query an Amazon RDS for PostgreSQL database in the data account. The VPCs have overlapping CIDRs and VPC peering is not permitted.
>
> The database has no load balancer in front of it, and its IP address changes on failover.

Which solution provides private access with the LEAST operational overhead?

A. A Network Load Balancer targeting the database's IP, exposed with PrivateLink, plus a Lambda function to update the target on failover

B. A VPC Lattice resource configuration for the database ARN, with a resource gateway, shared through AWS RAM

C. A Transit Gateway between the two accounts

D. Make the database publicly accessible and restrict it with a security group

◆ PAUSE AND DECIDE

Which solution provides private access with the LEAST operational overhead?

A NLB, PrivateLink and a Lambda

Works, and was the standard answer. The failover script is the overhead the question warns about.

B Lattice resource configuration

Identifies the database by ARN, so failover is handled. Shared privately across accounts.

C Transit Gateway

Overlapping CIDRs cannot be routed.

D Public database

Fails the requirement for private access.

THE SIGNAL

The changing IP on failover is planted to make the NLB answer expensive. An ARN-based configuration removes it.

What to carry out of this module

- **NLB** for fixed IPs, non-HTTP and PrivateLink. **ALB** for routing, auth and WAF.
- **REST API** for usage plans, WAF, caching and private endpoints. Otherwise HTTP API.
- **App Mesh is discontinued** as of 30 September 2026.
- **Service Connect** for ECS together. **Lattice** for any compute across VPCs and accounts.
- **Lattice resource configurations** reach databases and other TCP resources.

> ON THE EXAM
>
> Scope reversal: C02 courses call VPC Lattice out of scope. On C03 it is named in skill 1.1.7.